- The execution environment for code on the blockchain and rules for maintaining consistency, correctness, and immutability in this context

You get the idea. The structure of the blockchain and operations on it are not arbitrary, but well guided by a protocol. The implementation of the protocol establishes the base layer on which applications are written.

The framework for code execution introduced by the Ethereum blockchain protocol has opened a whole world of opportunities in the decentralized realm. The smart contract is the centerpiece and the main contribution of the Ethereum protocol.

Consider the stack diagram in figure 1.10, which compares the Bitcoin and Ethereum blockchains. The Bitcoin blockchain is for the transfer of cryptocurrency, and it does that job well. It has only wallet applications for initiating transactions. Ethereum supports smart contracts and a VM sandbox called Ethereum VM (EVM) on which the smart contracts execute. Smart contracts in turn enable decentralized operation of applications.